Adjustment to Vision Loss in a Mixed Sample of Adults with Established Visual Impairment

Daryl R. Tabrett and Keziah Latham

Purpose. To determine factors associated with the level of adjustment to vision loss in a cross-sectional sample of adults with mixed visual impairment.

Methods. One hundred participants were administered the Acceptance and Self-Worth Adjustment Scale (AS-WAS) to assess adjustment to vision loss. The severity of vision loss was determined using binocular clinical visual function assessments including visual acuity, contrast sensitivity, reading performance, and visual fields. Key demographics including age, duration of visual impairment, general health, education, and living arrangements were evaluated, as were self-reported vision-related activity limitation (VRAL), depression, social support, and personality.

Results. Multivariate analysis showed that higher levels of depressive symptoms ($\beta = -0.26$, $P < 0.01$) and of the personality trait neuroticism ($\beta = -0.33$, $P < 0.001$), and lower levels of the personality trait of conscientiousness ($\beta = 0.29$, $P < 0.01$), were associated with poorer adjustment to vision loss, explaining 56% variance.

Conclusions. Adjustment to vision loss is significantly associated with depression and certain traits of personality (specifically neuroticism and conscientiousness), independent of the severity of vision loss, VRAL, and duration of vision loss. The results suggest certain individuals may be predisposed to exhibiting less adjustment to vision loss due to personality characteristics, and exhibit poorer adjustment owing to or as a consequence of depression, rather than due to other factors such as the onset and severity of visual impairment. (*Invest Ophthalmol Vis Sci.* 2012;53:7227–7234) DOI:10.1167/iovs.12-10404

Adjustment to vision loss has been previously described as "accommodating to a vision loss by changing one's self-concept and goals to include the realistic restrictions that are imposed because of vision loss, while developing new capabilities that are compatible with personal resources."[1,2] It is thought to follow after the initial stages of mourning and bereavement associated with the loss of vision,[3] and may be a continual process rather than one with a definite endpoint.[2,4]

The psychosocial burdens that individuals experience when faced with vision loss are marked, and have been associated with increased mortality.[5–7] These psychosocial burdens associated with visual impairment may not pass and can persist, even after receiving low vision rehabilitation.[4,8] However, people vary greatly in their reactions to vision loss, likely being influenced by personal characteristics and social circumstance.[9] Therefore, difficulties have arisen in defining, measuring, and determining the significant factors involved in the process of adjustment to vision loss.[2,10,11]

The 19-item Acceptance and Self-Worth Adjustment Scale (AS-WAS) was developed to assess the unidimensional latent construct of adjustment to vision loss. The scale includes aspects of self-esteem, acceptance, locus of control, attitudes, and self-efficacy.[11] In addition to these factors, although not included in the AS-WAS itself, depression appears to hold significance in terms of adjustment, with lower adjustment expected in the presence of higher levels of depression.[3,12,13] Furthermore, worse adjustment has been related to a greater severity of vision loss,[12,14] but only due to the functional limitations vision loss causes rather than with the actual degree of visual impairment.[15,16] However, despite knowledge that people with greater depression and more functional limitations are likely to be less well adjusted,[12,15] other significant predictors of adjustment, as defined here, are currently unknown.

Personality may play a role in situations such as adjusting to vision loss because it is thought to affect the ability of an individual to cope in a flexible and constructive way,[17] including the extent to which positive reappraisal and planful problem solving is used.[18,19] In addition, personality is thought to determine the extent of any negative effect one may experience[20,21] during such situations as vision loss.[22] Social support is generally thought to be of importance when an individual is faced with a source of stress or period of transition[23] such as visual impairment and can act as a protective buffer to the distress of such events.[24] Social support can therefore have a positive effect on adjustment,[2,12,16,25] but may also be a hindrance[12,26–29] if negative support or overprotection occurs. Other proposed influences on adjustment include an individual's age, general health, living arrangements, and education.[12,16]

Given the recent increased emphasis on the provision of emotional support for people with visual impairment,[30,31] determining the factors associated with adjustment and their relative roles will help to further understand the process of adjustment to vision loss. Identifying such factors will help practitioners to recognize patients who may experience psychosocial burdens as a result of reduced adjustment as well as helping to spot any possible barriers to the successful implementation of interventions during low vision rehabilitation. The aim of this study, therefore, was to evaluate the factors associated with adjustment to vision loss in a mixed

From the Department of Vision and Hearing Sciences, Anglia Ruskin University, Cambridge, United Kingdom; and Vision & Eye Research Unit, Anglia Ruskin University, Cambridge, United Kingdom.

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Corresponding author: Keziah Latham, Department of Vision and Hearing Sciences, Anglia Ruskin University, East Road, Cambridge, CB1 1PT, UK; keziah.latham@anglia.ac.uk.

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sample of adults with established visual impairment while considering the severity of vision loss using an extensive assessment of visual function, to better understand the process of adjustment and to help identify individuals who are at risk of experiencing lower levels of adjustment.

Methods

Participants

One hundred individuals participated in the study, which was performed at Essex County and Clacton and District Hospitals and Anglia Ruskin University Eye Clinic. Participants at each location were approached, after having attended low vision support and rehabilitation sessions, and were included if they had experienced vision loss for more than 6 months that they felt caused restriction in their daily life. Those who were under 18 years of age, were unable to perform verbal evaluations in English, had no perception of light in both eyes, or were cognitively impaired (as determined by the Mini-Mental State Examination)[32] were excluded. Ethical approval was granted by Anglia Ruskin University Research Ethics Committee and NHS Essex Ethics Committee. The study complied with the tenets of the Declaration of Helsinki, and all participants gave informed consent after the nature and possible consequences of the study were explained. All study interviews and assessments were performed by the same examiner, a qualified optometrist (DRT), usually on the same day, but occasionally in two appointments within 2 weeks of each other. Participation was at least 2 weeks after patients' routine low vision assessments, which were not conducted by DRT.

Demographics

A structured face-to-face interview elicited key demographics including age, sex, primary cause of visual impairment, and the length of time since ocular diagnosis for each eye, living arrangements, and education. The presence of any comorbidity from a list of 12 common medical conditions was recorded and used to represent general health.[33]

Adjustment to Vision Loss

Adjustment to vision loss was assessed using the 19-item AS-WAS,[11] with each item graded on a four-point rating scale of strongly agree, agree, disagree, and strongly disagree. The AS-WAS evaluates the aspects of adjustment concerned with self-esteem, attitudes, locus of control, self-efficacy, and acceptance (Table 1). The AS-WAS is suitably unidimensional (first contrast eigenvalue 1.9; ideally <2.0) with good item fit (mean-square [MNSQ] infit 1.01 ± 0.11; outfit 0.99 ± 0.15; ideally 1.00) and person (2.38) and item (5.32) reliability coefficients (both ideally >2.00, i.e., reliability >0.80), good convergent and discriminant validity, and is free from systematic differential item functioning.[11,15]

Psychosocial Factors and Self-Reported Activity Limitation

Depressive symptoms were assessed with the 15-item Geriatric Depression Scale (GDS).[34] Questions are answered on a dichotomous response scale of yes or no. Although originally designed to be used in the elderly, the GDS has also been validated in younger samples.[35,36] The GDS was chosen primarily based on its previously demonstrated favorable psychometric properties,[37] including evaluations using item response theory[38,39] and its wide use in samples with visual impairment (e.g., Tolman et al.[13]; Brody et al.[40]; Rovner et al.[41]). Within the current sample the GDS was sufficiently unidimensional (first contrast eigenvalue 1.80), demonstrating acceptable fit statistics (MNSQ infit 0.98 ± 0.19, outfit 0.99 ± 0.38) and had good item (3.48) but not person (1.11) reliability.[15]

Table 1. The 19-Item Acceptance and Self-Worth Adjustment Scale (AS-WAS)

On the whole, I am satisfied with myself (E) (+)
At times I think I am no good at all (E) (−)
I am able to do things as well as most other people (E) (+)
I certainly feel useless at times (E) (−)
I feel that I do not have much to be proud of (E) (−)
I feel that I am a person of worth, at least on an equal plane with others (E) (+)
I wish I could have more respect for myself (E) (−)
I take a positive attitude toward myself (E) (+)
People with my sort of (visual) problem are constantly worried about what might happen to them (AT) (−)
People with my sort of (visual) problem feel that they are worthless (AT) (−)
It's what I can do to help myself that's really going to make the difference (LC) (+)
It's up to me to make sure I make the best of my future in these circumstances (LC) (+)
It makes me feel very bad to see all the things others can do that I cannot (AC) (−)
Because of my (visual) problems, I have little to offer other people (AC) (−)
In spite of my (visual) problems I feel satisfied with my abilities (AC) (+)
In just about everything, my (visual) problems are so annoying that I can't enjoy anything (AC) (−)
I give up on things before completing things (SE) (−)
When I make plans, I am certain I can make them work (SE) (+)
I do not seem to be capable of dealing with most problems that come up in life (SE) (−)

Items key: (E), esteem; (AT), attitudes; (LC), locus of control; (AC), acceptance; (SE), self-efficacy; (+) positively phrased; (−) negatively phrased.

Personality was assessed according to the five-factor model using the Neuroticism, Extraversion, Openness Five Factor Inventory (NEO FFI).[21] The NEO FFI contains 60 statements relating to the traits of neuroticism, extraversion, agreeableness, conscientiousness, and openness to experience. Each item is rated on a five-point response scale of strongly disagree, disagree, neutral, agree, and strongly agree.

Neuroticism relates to the likelihood of being tense, insecure, self-conscious, and more prone to experiencing negative emotions such as depression and anxiety.[21,42] Extraverts are sociable and assertive, and more likely to experience positive emotions.[20,21,43,44] Openness to experience represents a higher level of imagination and curiosity, and those with this trait are more likely to be flexible in their thinking, experiencing a diversity of emotions.[21,44] Individuals higher in agreeableness tend to be helpful and trusting, and higher conscientiousness represents greater organization, self-discipline, and dependability.[20,21,44] The neuroticism, extraversion, openness, agreeableness, and conscientiousness subscales were largely unidimensional (first contrast eigenvalues between 2.0 and 2.3), demonstrating good fit statistics (mean MNSQ infit between 0.99 and 1.02; outfit between 0.99 and 1.03) and showed fair-to-good person reliability (1.45–2.61) and excellent item reliability (3.24–7.15).[15]

Social support was assessed using the 12-item Interpersonal Support Evaluation List (ISEL).[45] The ISEL is designed to assess perceived available support in three areas: appraisal (four statements), belonging (four statements), and tangible support (four statements). Each item is rated on a four-point response scale of definitely true, probably true, probably false, and definitely false. The ISEL was adequately unidimensional (first contrast eigenvalue 2.1), demonstrated good item fit statistics (MNSQ infit 1.01 ± 0.14, outfit 0.98 ± 0.15), and had good item (2.80) and fair person (1.93) reliability.[15]

Self-reported vision-related activity limitation (VRAL) was assessed using a refined version[15] of the Activity Inventory (AI),[46] which contains a wide range of daily activities specifically chosen to represent the reading, mobility, visual information, and visual motor abilities of a population with mixed visual impairment. Some goals of the original AI that were less relevant in the United Kingdom and deemed less essential in the current sample were not administered (e.g., hunting and shooting). The instrument implemented contained 30 goals and 235 tasks across three objectives (the items included can be seen elsewhere[15]). The importance of each goal was rated on a four-point scale (not important, slightly important, moderately important, and very important), and the difficulty of important goals and the tasks underpinning these goals were then graded by the respondent on a five-point scale (not difficult, slightly difficult, moderately difficult, very difficult, and impossible without assistance). Adaptive testing was used in that only the difficulty of tasks serving important and difficult goals was assessed, and these responses were used to formulate overall self-reported VRAL at task level which was used for analysis. The use of Rasch analysis means that the difficulty of every item does not have to be graded to formulate an accurate estimate of overall self-reported VRAL.

Clinical Visual Function

For clinical visual function assessments participants wore any habitual distance spectacle prescription (and a corresponding reading addition where necessary):

Distance visual acuity was measured binocularly with an externally illuminated Bailey–Lovie chart[47] at 3 m (chart luminance, 95–100 cd/m^2) assessed on a per letter basis until no letters on a line could be correctly identified.[48,49]

Contrast sensitivity was measured binocularly with a Pelli–Robson chart[50] at 1 m (chart luminance, 95–110 cd/m^2) assessed per letter until no letters in a triplet could be correctly identified.[51–53]

Near reading performance was measured using an MNRead chart at 40 cm (moved closer in a log unit progression if necessary) (chart luminance, 110–150 cd/m^2).[54,55] The maximum reading speed, critical print size, and near reading acuity were derived for analysis.[56]

Static threshold binocular central visual fields were conducted with a visual field analyzer (Humphrey Field Analyzer; Carl Zeiss Meditec, Inc., Dublin, CA), using a Central 30-2 SITA-Fast strategy.[57] The mean threshold of the central 30 degrees was calculated and used for analysis.

Statistical Analysis

All data were double entered and corrected for errors before data analysis. Rasch analysis was conducted using Rasch analysis software (Winsteps version 3.69.1; WINSTEPS, Chicago, IL) on responses to all self-report instruments to provide person measure estimates for each construct. Univariate analyses were first undertaken to explore the demographic, visual function, self-report, and psychosocial variables. Bivariate analyses in the form of linear correlation coefficients were then conducted to investigate the association between the predictor variables and adjustment. Finally, hierarchical multiple regression was conducted to investigate the relationship between multiple predictor variables and adjustment.

For the multiple regression analysis, residuals were normally distributed and the assumptions of homoscedasticity (the extent to which the variance of residuals was equal for all predicted values) and linearity were supported. The variation inflation factor was sufficiently low for all variables and the intercorrelations among the independent variables within the regression analysis were acceptable ($r \leq 0.60$), both indicating absence of bias from multicollinearity. The standardized residuals supported adequate fit of the sample to the model.

Results

Table 2 summarizes the descriptive characteristics of the study sample. The range of AS-WAS scores was approximately 4 logits, reflecting a wide range of adjustment levels in the current sample. All personality traits were within 0.6 SDs of the general population.[21] Linear correlation coefficients revealed that none of the assessed demographic variables (age, length of time with visual impairment, education, living arrangements, and number of comorbidities) was significantly associated with the level of adjustment to vision loss ($P > 0.05$ in all cases). In contrast, all measures of clinical visual function (except distance visual acuity and maximum reading speed), self-reported VRAL, and psychosocial variables (except the personality trait openness) were significantly associated with the level of adjustment (see Table 3).

Since previous studies have shown a significant relationship with adjustment,[12,13,15] self-reported VRAL and depression were entered in step 1 of a hierarchical regression analysis. Significant measures of clinical visual function identified in bivariate analyses were force entered in step 2 to confirm the relationship with adjustment when self-reported VRAL is considered,[15] and all other variables significantly associated with adjustment in bivariate analyses were added in a final exploratory step in a stepwise manner (Table 4). It can be seen from Table 4 that in step 2 of the regression, once depression, VRAL, and severity of visual loss are considered together, severity of visual loss and VRAL are no longer significant predictors of the level of adjustment. Depression itself is associated with functional limitations but not with the severity of vision loss.[15] In step 3, social support ($P = 0.25$) and the personality traits of extraversion ($P = 0.70$) and agreeableness ($P = 0.34$) are not selected as significant predictors, whereas the significance of neuroticism ($\beta = -0.33$, $P < 0.001$), conscientiousness ($\beta = 0.29$, $P < 0.01$), and depression ($\beta = -0.26$, $P < 0.01$) remain.

Discussion

The results indicate that better adjustment to vision loss is associated with lower levels of depression and with certain personality traits, specifically lower levels of neuroticism and greater conscientiousness (see Fig.). Furthermore, factors including severity of vision loss, self-reported VRAL, social support, extraversion, and agreeableness, although significantly associated with adjustment in bivariate correlations, are not important when considered in multivariate analysis. The study also identifies that other notable variables such as the length of time with vision loss, general health, living arrangements, education, and age are not associated with the degree of adjustment in a cross-sectional sample of adults with established mixed visual impairment. Despite the results of the current study indicating that adjustment to vision loss is not associated with the severity of visual impairment, similar to the findings of some[16] but not others,[12,14] note that all participants had some degree of residual vision, and so it is unclear whether this would also hold for individuals with no perception of light (i.e., total blindness).

The importance of depression with regard to adjusting to vision loss has been recognized previously. Better adjustment as measured by the Adaptation to age-related Vision Loss scale (AVL) was significantly correlated with fewer depressive symptoms in an elderly sample of people with mixed visual impairments[12] and in individuals with age-related macular degeneration (AMD).[13] Results of the current study add to these findings by indicating that depression is associated with poorer adjustment to vision loss, independently of the severity

Table 2. Descriptive Statistics of the Study Sample ($n = 100$)

Sex, %		39 male, 61 female	
Primary ocular diagnosis, %	Right eye		Left eye
Macular dysfunction (including AMD)	54		57
Optic neuropathy (including glaucoma)	11		11
Diabetic retinopathy	7		8
Retinal (including myopic degeneration)	5		5
Undiagnosed	8		9
Cataract	2		2
Corneal	3		2
Other	10		6
Education, %			
Standard school leaving age		59	
A-level or equivalent		13	
Further education		12	
Higher education		2	
Bachelors degree		10	
Postgraduate qualification		4	
Living arrangements, %			
Alone		40	
With partner		51	
With other		6	
Warden assisted		3	
	Average		Min–Max
Age, y	81 (73–86)		25 to 98
Time since most recent 1° diagnosis, y	5 (3–12)		0.54 to 64
Number of comorbidities	1.5 (1–3)		0 to 6
Adjustment to vision loss (AS-WAS; logits)	1.53 (0.84–1.93)		0.71 to 4.69
Distance visual acuity, logMAR	0.84 ± 0.36		0.08 to 1.64
Contrast sensitivity, logCS	1.10 (0.88–1.35)		0.15 to 1.75
Near reading performance			
Reading acuity, logMAR	0.81 ± 0.45		−0.04 to 2.00
CPS, logMAR	1.00 (0.80–1.30)		0.10 to 2.00
Maximum reading speed, words/min	107.36 (45.95–143.7)		1.34 to 210.52
Visual field, dB	22.28 (14.21–26.40)		0.11 to 30.32
Self-reported VRAL (AI; logits)	1.30 ± 1.23		−2.31 to 5.20
Depression (GDS)			
Raw score	2 (1–4)		0 to 14
Logits	−2.40 (−3.36 to −1.26)		−4.77 to 3.15
Social support, ISEL			
Raw score	21 (17–25)		12 to 40
Logits	1.28 ± 1.07		−1.37 to 4.96
Personality (NEO FFI)			
Neuroticism (40%)*			
Raw score	13.50 (9–21)		0 to 45
Logits	−0.84 (−1.23 to −0.25)		−5.15 to 2.80
Extraversion (47%)*			
Raw score	26.86 ± 6.25		14 to 40
Logits	0.33 ± 0.67		−1.05 to 1.93
Openness (39%)*			
Raw score	24.87 ± 6.72		8 to 39
Logits	0.05 ± 0.52		−1.43 to 1.30
Agreeableness (79%)*			
Raw score	36 (32–40)		20 to 47
Logits	1.39 ± 0.73		−0.25 to 4.31
Conscientiousness (62%)*			
Raw score	36.01 ± 6.36		20 to 47
Logits	1.23 (0.70–2.19)		−0.44 to 4.72

The mean ± SD is provided for the variables whose frequency distributions were not significantly different from normal. For all others, the median (25%–75% interquartiles) is given.

* Percentile equivalents: 50% = normal data average for a given trait; <50% = sample lower in trait; >50% = sample higher in trait, compared with normal data (Costa and McCrae, 1992[21]).

of vision loss (as assessed by a range of clinical visual function tests) and the degree of functional limitations. Therefore, patients with greater depressive symptoms are more likely to be less well adjusted irrespective of the degree of visual impairment and the functional limitations that vision loss causes. It should be acknowledged, however, that depression itself in these patients can lead directly and reciprocally to functional limitations and disability.[58] Taken as a whole, it is

Table 3. Linear Correlation Coefficients (2-Tailed Spearman's) for the Associations between Predictor Variables and Adjustment to Vision Loss (AS-WAS Score, logits)

	r
Demographics	
Sex	−0.08
Age	−0.03
Time since diagnosis	0.03
Education	−0.02
Living arrangement	−0.05
Number of comorbidities	−0.08
Self-reported VRAL	0.46***
Depression	−0.54***
Social support	0.35***
Personality	
Neuroticism	−0.65***
Extraversion	0.37***
Openness	−0.03
Agreeableness	0.26**
Conscientiousness	0.29**
Severity of vision loss	
Distance visual acuity	−0.16
Contrast sensitivity	0.25*
Near reading acuity	−0.21*
Near CPS	−0.20*
Maximum reading speed	0.14
Visual field	0.24*

Due to the number of comparisons (21) it should be noted that approximately one (21 × 0.05) correlation would be expected to be significant by chance alone at the $P = 0.05$ level.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

Table 4. Results of Hierarchical Multiple Regression Analysis for Predicting Adjustment to Vision Loss

	B	SE B	β
Step 1 ($R^2 = 34.1\%$)			
Constant	0.69	0.14	
Self-reported VRAL	0.19	0.08	0.23*
Depression	−0.28	0.06	−0.44***
Step 2 ($R^2 = 35.6\%$)			
Constant	0.03	0.58	
Self-reported VRAL	0.13	0.11	0.16
Depression	−0.29	0.06	−0.46***
Severity of vision loss			
Contrast sensitivity	0.15	0.36	0.05
Near reading acuity	−0.46	0.57	−0.22
Near CPS	0.66	0.65	0.26
Visual fields	0.01	0.01	0.09
Step 3 ($R^2 = 56.3\%$)			
Constant	−0.46	0.52	
Self-reported VRAL	0.41	0.10	0.17
Depression	−0.17	0.06	−0.26**
Severity of vision loss			
Contrast sensitivity	0.43	0.31	0.15
Near reading acuity	−0.27	0.48	−0.13
Near CPS	0.58	0.55	0.23
Visual fields	−0.00	0.01	−0.01
Additional factors			
Neuroticism	−0.26	0.07	−0.33***
Conscientiousness	0.25	0.07	0.29**

Self-reported VRAL and depression were force entered in step 1, then clinical visual function force entered in step 2, and social support and personality were entered stepwise in step 3.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

suggested that the domain-specific aspect of psychological well-being that is adjustment to vision loss may contribute directly to the more global measure of depression and, jointly, depression may lead directly to poorer adjustment in terms of factors such as reduced self-esteem and self-efficacy as assessed by the AS-WAS.

According to the five-factor model of personality structured by the NEO FFI, it is anticipated that an individual will behave in a similar way given any situation or circumstance, given that personality is thought to be relatively stable throughout one's lifetime and is partly a result of genetic predisposition.[59,60] The results suggest that those individuals with higher neuroticism

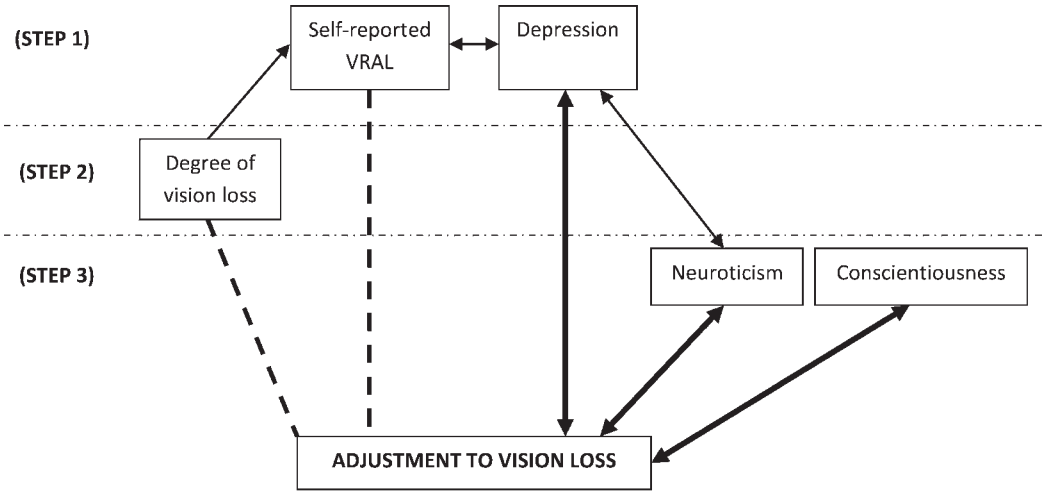


Figure. The conceptual framework of the factors associated with adjustment to vision loss. Steps 1, 2, and 3 refer to the hierarchical regression analysis presented by the current study. The *bold solid arrows* represent significance between variables after step 3 of the hierarchical regression analysis. The *dotted lines* between variables represent the loss of a significant association (found initially including in bivariate correlations) after subsequent steps of the hierarchical regression analysis. The *nonbold solid arrows* represent significance between variables found in previous research.

and/or lower conscientiousness will be more prone to experience difficulties in adjustment when faced with vision loss. Similar to depression, the association between personality and adjustment appears to occur regardless of the severity of vision loss and functional limitations.

Neuroticism relates to the likelihood of being tense, insecure, self-conscious, and more prone to experiencing negative emotions such as depression and anxiety.[21,42] Therefore, it is perhaps not surprising that higher neuroticism is significantly associated with lower adjustment, considering that the AS-WAS assesses areas of adjustment relating to confidence and motivation, including self-esteem and self-efficacy. In addition, it has been proposed that those high in neuroticism can also appear "to have problems in coping in constructive ways" and "may be inflexible in their coping,"[17] which could potentially hinder adjustment to visual impairment.

Higher conscientiousness represents more organization, self-discipline, dependability, competence, and a greater striving for achievement.[20,21,44] Individuals high in conscientiousness have been shown to use positive reappraisal and problem-solving techniques when coping with stressors,[19] to be more likely to seek support and adapt to the demands of a situation, and, therefore, to respond to challenges in appropriate ways.[17] These advantageous attributes can help explain the significant relationship between conscientiousness and adjustment as measured by the AS-WAS, and could represent a greater determination to seek, to proceed, and to be successful with different rehabilitation strategies used in the adjustment process.

Although personality is thought to be relatively stable throughout one's lifetime, aspects of adjustment such as self-esteem and self-efficacy, like depression, are dynamic and can change over time and with situation. Therefore, although personality traits may lead to certain individuals being more prone to having difficulty adjusting when faced with vision loss, perceptions and moods can be altered with training and rehabilitation.[61] Although people can vary enormously in their reactions to vision loss[62,63] and, although causality is not known, results of the current study suggest that addressing depression, adjustment, and functional limitations are key to successful low vision rehabilitation.

Different treatment methods have demonstrated a positive effect in improving adjustment, depression, and functional limitations, although there are no validated protocols.[64] A 6-week 12-hour AMD cognitive education program has been seen to lower the incidence of depression, reduce the severity of symptoms among depressed patients, increase self-efficacy, and improve self-reported visual function.[65–68] Undertaking routine low vision rehabilitation can improve adjustment to vision loss,[69] as can teaching functional living skills including orientation and mobility training and occupational therapy for at least 4 weeks.[70] Problem Solving Treatment (PST)[22] may be of particular benefit, in that the treatment addresses individuals' negative perceptions that may interfere with finding practical solutions to difficulties in daily life and teaches problem-solving skills,[71] potentially addressing some of the particular issues faced by those with high neuroticism and low conscientiousness. PST has been shown to be effective in preventing depression in the short term in people with macular disease.[72] There may also be benefits in adjustment, depression, and reported visual function by using group instead of one-to-one interventions.[73]

The lack of relationship between adjustment and the duration of visual impairment found in the current study suggests that implementation of rehabilitation should not be confined to soon after diagnosis but may be beneficial at any time in the course of vision loss. The need for help in adjusting to vision loss is more dependent on the personality of the patient than the duration of the vision loss.

The range of adjustment to vision loss in the current sample is wide (approximately 4 logits), and could be taken to indicate that adjustment can be seen as a continuous process without a definite endpoint. However, the results are cross-sectional and reflect individuals' adjustment at a fixed point in time, relative presumably to their psychological state prior to becoming visually impaired. Therefore, we cannot comment on how individuals adjust to vision loss over time, and further research is needed to investigate individuals' adjustment profile longitudinally, to improve understanding of the pathway of adjustment, and to empirically determine the consequences of poor adjustment in the rehabilitation process.

In summary, the results of the present study show that depression and personality (neuroticism and conscientiousness) are significantly associated with adjustment to vision loss independent of the duration of visual impairment or the severity of visual loss as assessed by self-reported VRAL and a range of clinical visual function tests.

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